Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		2624793

Pacemaker: A continuous time model for accurate prediction and analysis of battery status

Summary

Batteries are as important as the heart. In 1786, the twitching of the frog on the Galvanic dissecting table illuminated the birth of the Voltaic stack. As Tesla once said, 'If you want to explore the mysteries of the universe, you must think from the perspectives of energy, frequency, and vibration.' Batteries are the core carrier for humans to decode energy. It ranges from driving mobile phones and pacemakers to empowering new energy vehicles and energy storage power stations, breaking down energy barriers in time and space. Thus, the battery model, like a pacemaker, maintains people's research on batteries. **This paper aims to establish a quantitative model to achieve accurate battery charging, power consumption prediction, and in-depth analysis.**

For Task I, we have developed a totally new model called **THMSVM** to describe the charging state of the battery. This model belongs to a continuous time system and **couples multidimensional factors** such as temperature and SOH decay on the traditional first-order Thevenin model. We use **PINN** for forward solving of the model and **MCPSO** for reverse parameter identification. We validated the terminal voltage and polarization voltage of the model, and used **LSTM-Attention-AKF** to estimate SOC. **Multi-feature fusion LSTM Attention** was used to predict SOH, and the prediction results were excellent.

For task II, we adopt **Markov modulation Monte Carlo method** for TTE modeling, which integrates SOH and random load to enhance the confidence of the model. In addition, we used the **Radau IIA implicit Runge Kutta method** to solve the model and quantify the uncertainty. Verified the power prediction under different SOC, usage scenarios, mixed SOC and usage scenarios, and SOH, fully evaluated the performance of the model, and identified the driving factors for battery depletion in each situation.

For task III, we use the single factor perturbation method (**OAT**) for local sensitivity testing; Using Latin Hypercube Sampling (**LHS**) mixed with Random Forest Forward (**RF**) for global sensitivity analysis, quantifying the contribution of parameters in the entire space. And analyze and predict the fluctuations under three scenarios: modeling assumptions, parameter values, and usage patterns.

For task IV, we analyze the mechanism by which SOH decay accelerates depletion rate from a physicochemical perspective. And provide reasonable suggestions from both users and developers, finally return to the essence of THMSVM and extend it to other portable devices with **thermoelectric coupling** characteristics, demonstrating the high flexibility and portability of the model.

Keywords: Improved Thevenin model; PINN; MCPSO; LSTM-Attention-AKF; Random Forest

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1 Introduction

1.1 Background and Literature Review

Lithium ion batteries supply power for all modern smart phones, but its discharge law is difficult to predict. Different from the linear depletion of the traditional fuel tank, its electrochemical system is nonlinear, and the running time is significantly different under the same use mode. Unpredictability comes from the instantaneous power consumption such as display brightness and calculation intensity, as well as the background energy consumption when the equipment is idle.

In the long run, irreversible chemical processes will degrade battery performance, including lithium loss caused by SEI growth, electrode degradation, electrolyte decomposition and internal resistance increase. The state of charge (SOC) cannot be directly measured, and it needs to be inferred from voltage, current, temperature and use history. However, the power consumption of mobile phones fluctuates between milliwatts and seven watts, which increases the difficulty of estimation.

The high and low temperatures temporarily reduce the capacity and accelerate the degradation, and the short-term effects are superimposed with the long-term aging, so that the power consumption of the equipment used for 2 years may be twice that of the new equipment. To understand this dual time scale behavior, it is necessary to establish a model to connect the instantaneous electrochemical kinetics and long-term degradation process within the framework of actual mobile phone operation.

1.2 Restatement of the Problem

We are required to address the following problems through in-depth analysis and research combined with the specific constraints given.

Task 1: Develop a continuous time battery model that couples external factors and internal module power consumption.

Task 2: Based on the first battery model, conduct TTE prediction, analysis, evaluation of the model, and self consistently explain the role of model parameters.

Task 3: Sensitivity testing and analysis of the impact of changes in intrinsic factors of which models on model predictions.

Task 4: Implement the battery model project, provide suggestions for extending TTE from the user end and operation stage, and consider the portability of the model.

1.3 Our Work

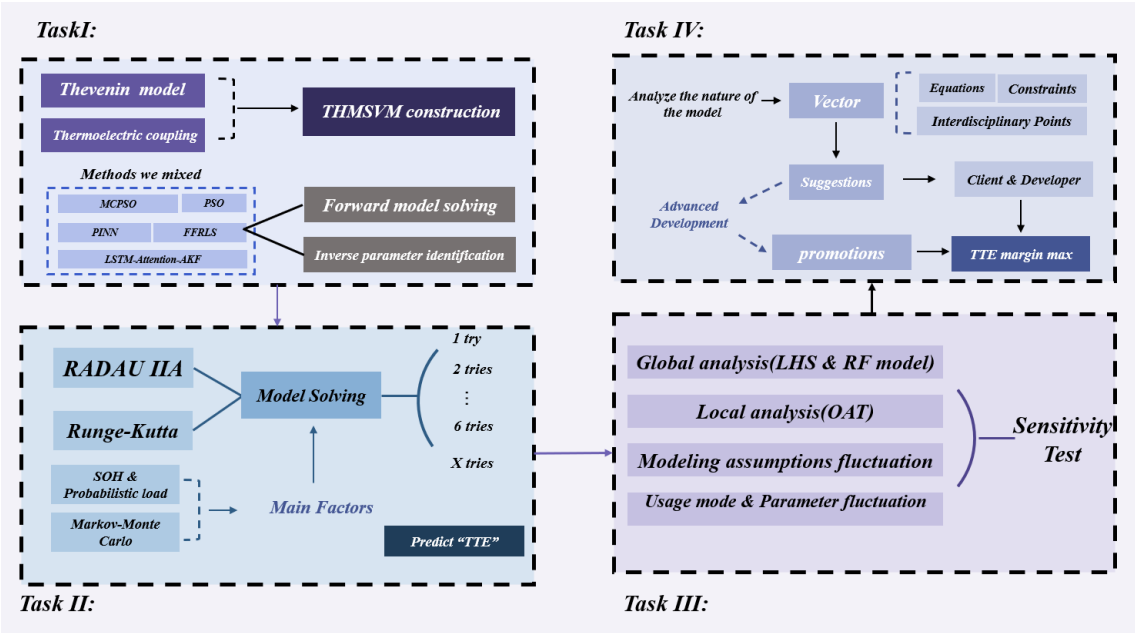

Figure 1: Flow chart of our work

2 Assumptions and Justifications

We develop several assumptions considering the conditions required for modeling:

Assumption 1: The lithium-ion battery is represented by **a continuous-time Thevenin equivalent circuit model**.

Justification: The complex charge transfer and diffusion processes within the battery can be effectively characterized by a single RC parallel network. Models fitted with discrete data do not have universality in the physical world.

Assumption 2: The smartphone operates as **a Constant Power Load** within each discrete simulation time step (Δt).

Justification: Modern mobile terminals are typically equipped with PMIC. In order to maintain the power supply demand of the components, the current will automatically increase as the battery voltage decreases. Therefore, the load is modeled as a constant power, rather than a constant current or constant resistance.

Assumption 3: The variation of battery internal resistance with temperature strictly follows the **Arrhenius equation.**

Justification: This assumption excludes secondary factors such as pressure and mechanical deformation, emphasizing that reduced ionic activity at low temperatures induces an exponential increase in internal resistance, which constitutes the primary physical mechanism for voltage drop.

3 Notations

The primary notations used in this paper are listed in Table 1.

Table 1: Notations

Symbol	Description
$SOC(t)$	state of charge with time
$U_{OCV}(SOC)$	open-circuit voltage
P_{sum}	total consumption
P_{se}	screen brightness & standby mode power consumption
P_{cpu}	processor power consumption
P_{bg}	background task power consumption
P_{nt}	network induced consumption

4 Task I: Developing battery continuous time model

4.1 Problem analysis & strategy design

To fulfill the specified requirements, a continuous physics-based battery model was developed to accurately capture the dynamic charging and discharging behaviors of lithium-ion batteries under real-world operating conditions. Discrete or purely empirical modeling approaches—such as those based on training data fitting or deep learning—often lack robustness in independent real-world deployment, thereby limiting their engineering applicability and impeding commercialization and broad adoption.

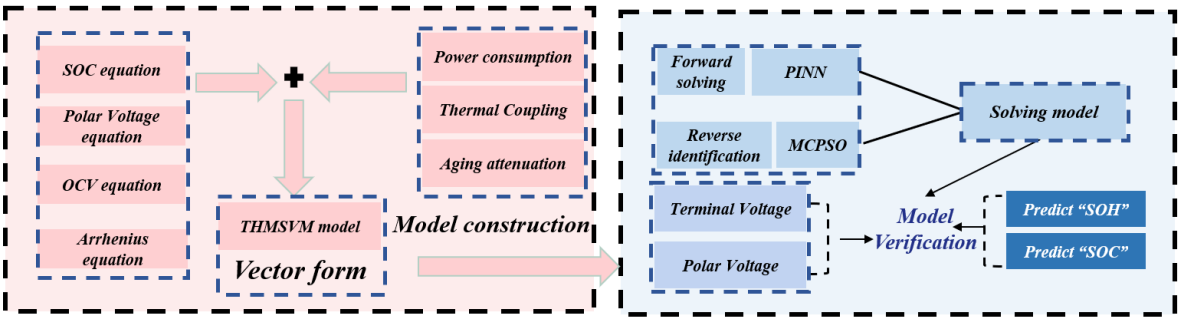

Figure 2: Problem solving flowchart for Task I

In light of these limitations, an extensive review of existing battery modeling literature was conducted. While various modeling frameworks exhibit distinct advantages, none were found to fully satisfy the required design criteria. Consequently, **a novel continuous-time model, designated THMSVM, was independently developed.** Compared to electrochemical models such as the pseudo-two-dimensional (P2D) and single-particle (SPM) models, THMSVM entails significantly lower computational resource demands and reduced processing overhead. Moreover, in contrast to conventional equivalent circuit models—including the Rint, Thevenin, and second-order RC configurations—THMSVM delivers superior estimation accuracy and enhanced real-time performance, achieving an **optimal trade-off between computational efficiency and modeling fidelity.**

The principal innovation of THMSVM lies in its extension of the classical and highly

reproducible Thevenin equivalent circuit model through the coupling of three critical state variables: state of charge (SOC), polarization voltage, and battery temperature (T), thereby forming a three-dimensional state vector. To accommodate diverse operational scenarios, a detailed analysis of the power consumption characteristics associated with various smartphone subsystems was performed. This analysis informed the development of an energy consumption function and the subsequent estimation of calibration parameters, with all parameter identification conducted using dataset by NASA[1].

4.2 THMSVM development

The derivation of THMSVM can be divided into several stages, and each stage is shown in bold.

Stage 1: SOC equation

$$\frac{dSOC(t)}{dt} = -\frac{\gamma \cdot I(t)}{C_m \cdot 3600}$$

SOC is the percentage of the remaining battery capacity relative to the rated capacity, Where γ is the coulomb efficiency coefficient, C_m is the nominal capacity of the battery , $I(t)$ is the system control input, and the differential of SOC is proportional to it.

Stage 2: Polar voltage(V_p) equation

$$\frac{dV_p(t)}{dt} = -\frac{V_p(t)}{\tau} + \frac{I(t)}{C_p}$$

This equation describes the voltage change caused by the polarization effect during the charging and discharging process of the battery. Its dynamic characteristic parameters and transient response speed change with SOC, load mutation and temperature.

Stage 3: OCV equation

$$U_{OCV}(SOC) = \Gamma_0 + \Gamma_1 \cdot SOC + \Gamma_2 \cdot SOC^2 + \Gamma_3 \cdot SOC^3 + \frac{\Gamma_4}{SOC + \Gamma_5} + \Gamma_6 \cdot \ln(SOC) + \Gamma_7 \cdot \ln(1 - SOC)$$

This is an expression of approximation realized by the combination of polynomial, fractional and logarithmic terms, which is convenient for physical modeling and numerical solution.

Stage 4: Arrhenius Equation

$$R_t(t) = R_{rf} \exp[\xi(\frac{1}{t(\tau)} - \frac{1}{t_{rf}})]$$

Arrhenius formula is a classical formula to characterize the temperature change of internal resistance, which lays the foundation for our subsequent thermodynamic coupling.

Stage 5: Subdivision and coupling of power consumption

We classify the power consumption of different modules of mobile phones and express them with formulas respectively, and integrate them into the original SOC differential formula.

$$P_{nt}(t) = \sum_{a=1}^{6} P_{na}\delta(D(t) - d)$$

The above equation is the expression for $P_{nt}(t)$, it is worth noting that part of the energy consumption is treated approximately, that is, it is regarded as a constant. In network consumption, the number of items participating in the addition is set to 6, which means 6G, 5G, 4G, 3G, Wi Fi, Bluetooth to get more credible data calculation.

Stage 6: Thermal coupling

Effective capacity is a factor that cannot be ignored in battery use. With the accumulation of use time, the effective capacity of battery tends to decrease. This data is common in the whole series of Apple phones.

Therefore, based on the capacity data under typical temperature conditions such as 3°C, 22°C, 33°C and 43°C provided by NASA data set[1], combined with the temperature characteristics of lithium-ion batteries, the expression of dynamic power consumption term a is obtained:

$$\mathrm{P_{dy}(t) = m_n l(t) + k}r\mathrm{(t)^2 + q}r\mathrm{(t)} + \sum_{\mathrm{a=1}}^{6}\mathrm{P_{na}}\delta\mathrm{(D(t) - d) + P_b}$$

Stage 7: Aging attenuation coupling

This is very common. There are many factors causing battery aging, such as incorrect charging habits, accidental collision of mobile phones, professional titles of power management chip technology and circuit design, etc. To facilitate processing, we uniformly use the double exponential decay model to describe the aging coefficient, and integrate the time to obtain the expression of $A(T,N)$:

$$A(T,N) = \alpha(T) * \lambda(\int_0^t T_{cy}(t)^{-1} d\tau)$$

Stage 8: Multidimensional state vector modified SOC formula

Assemble a, B, C into a state vector x:

$$X = [SOC(t), t(\tau), Vp(t)]^{-1}$$

The fixed SOC formula is obtained by coupling X and all losses in the above assumptions:

$$\frac{dSOC(t)}{dt} = -\frac{\gamma * [P_{dy}(t) + P_b]}{C_m * V_{ref} * A(T,N)}$$

4.3 Solving the model

Part 1: Forward tackling

Considering that the battery model is a set of nonlinear rigid differential equations, after comparing the Radau IIA method, avos BDF method, EI method, PRK method, PS method and DG method based on implicit Runge Kutta, we adopt the physical information neural network direct solving method **(PINN)**.

PINN has the advantages of meshless discretization, no iterative solution, and strong physical interpretability. Compared with traditional numerical methods, it has significant advantages in scenarios such as battery life cycle simulation, aging battery performance prediction, and rapid evaluation of new battery systems.

The general form of the nonlinear rigid differential equations of the battery system can be expressed as:

$$\begin{cases} \dfrac{du(t)}{dt} = f(u(t);t;\theta), t \in [t_0, t_f] \\ u(t_0) = u_0 \end{cases}$$

Then we construct physical constraint loss function, the loss function L_{total} is defined as the weighted sum of the data loss item L_d, the physical loss item L_p and the initial condition loss item L_{ic}. The definition of each loss item is shown in the following table:

Weighting coefficient $\lambda_d, \lambda_p, \lambda_{ic}$ is used to balance the contribution of different constraints. The data loss term L_d represents the error between the predicted value of the neural network and the measured or simulated data, which is used to correct the model, such as battery aging, side reactions and unmodeled dynamics; The physical loss term L_p is used to ensure the physical consistency of the solution by forcing the neural network output to meet the control differential equation of the battery; The initial condition loss term L_{ic} forces the neural network output to meet the initial state constraint.

After that, PINN uses L-BFGS or Adam method to minimize the total loss function and update the parameters of neural network.

$$\{W^*, b^*\} = \arg\min_{W,b} L_{total}(W, b)$$

In the formula, W^* and b^* are the optimal parameters. When the loss function converges to the set threshold, the neural network $\mathbb{N}(t; W^*, b^*)$ is the approximate analytical solution of the battery differential equations.

Part 2: Reverse parameter identification

Inverse parameter identification is a process of estimating model parameters with known input and output data. Its mathematical expression is a constrained optimization problem:

$$\min_{\emptyset} J(\emptyset) = \sum_{a=1}^{N} [y_a - \hat{y}_a(\emptyset)]^2 \cdot w_a$$

Where ϕ is the parameter vector to be identified, J (ϕ) is the objective function, y_a is the measured output (terminal voltage) at the ith time point, $\hat{y}_a(\emptyset)$ is the output predicted by the model under the given parameter ϕ, w_a is the weight coefficient, n is the number of data points, ϕ min and ϕ max are the lower limit and upper limit of the parameter respectively.

We compared and analyzed the commonly used battery model parameter identification algorithms, and finally selected the **MCPSO** with stronger global optimization ability, better anti-interference robustness and wider adaptability to working conditions.

Part 2.1: Particle velocity and position update

Through master-slave group cooperation and information sharing, the balance between global exploration and local development is achieved to avoid falling into local optimization.

Speed update formula:

$$v_{i,d}^{t+1} = w * v_{i,d}^{t} + c_1 * r_1 * (p_{best} - x_{i,d}^{t}) + c_2 * r_2 * (g_{best} - x_{i,d}^{t}) + c_3 * r_3 * (m_{best} - x_{i,d}^{t})$$

Location update formula:

$$x_{i,d}^{t+1} = x_{i,d}^{t} + v_{i,d}^{t+1}$$

Where $x_{i,d}^{t}$ represents the position of particle i in the t-th iteration, and $v_{i,d}^{t+1}$ represents the updated speed.

Part 2.2: Fitness function calculation

In this stage, the error between the model output and the measured data is quantified to guide the particles to evolve in the direction of optimal parameters, which is the only basis for parameter selection.

$$f(fit) = \frac{1}{N} \sum_{a=1}^{N} (U_{\mathrm{mod}}(a) - U_{avg}(a))^2$$

As shown in the above formula, the optimization objective is to minimize $f(fit)$, so that the model output is highly consistent with the measured value.

Totally, MCPSO improves the optimization performance through master-slave group structure and information interaction. The following is the result of parameter identification.

After our calculations, we have obtained that the optimization range of some parameters is extremely large, which precisely confirms the accuracy and rationality of our parameter identification. For K_{ocv}, the change exceeds 300%, indicating that there is a significant difference between the initial parameter estimation and the actual battery characteristics. The optimized model can better reflect the actual behavior of the battery than the initial model.

4.4 Model Verification

To verify the accuracy of the model, we conducted tests on five variables, and further conducted a deeper correlation analysis for SOH.

Verification 1: Terminal Voltage & Polarization Voltage & Temperature

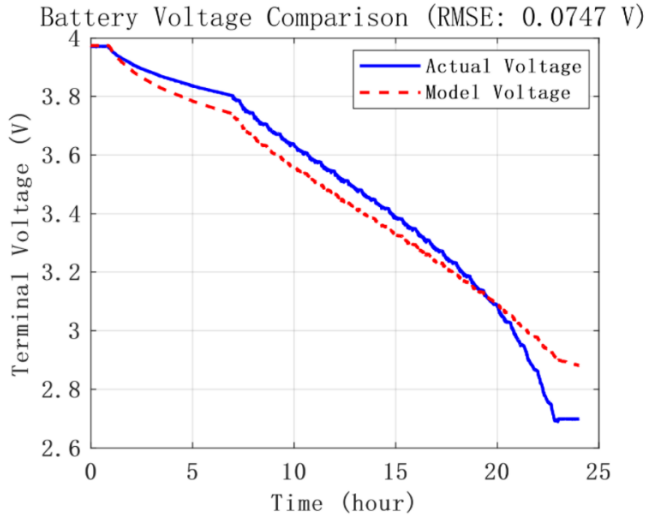

Figure 3: Analysis of thermoelectric voltage fitting effect

The root mean squared error (RMSE) of voltage prediction is approximately 0.0917V, indicating high prediction accuracy.

Verification 2: Polarization Voltage

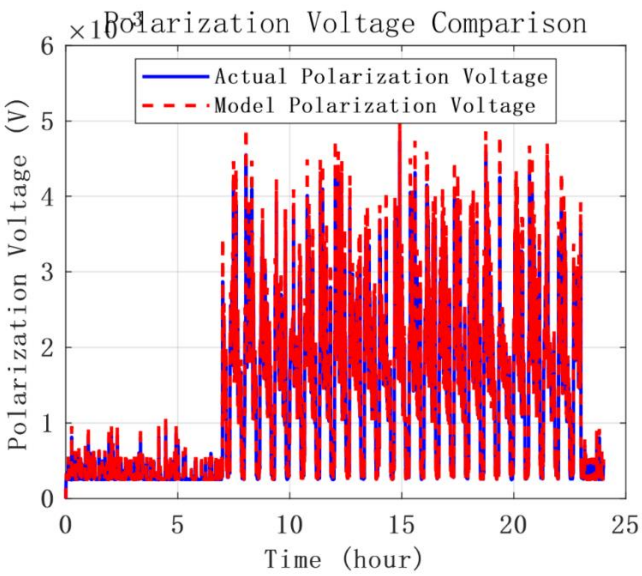

Figure 4: Analysis of polarization voltage fitting effect

As can be seen from the figure, the fitting results are relatively accurate across different time periods. However, as time accumulates and the influencing factors of polarization voltage increase, the fitting results become slightly less accurate.

Verification 3: SOC

SOC is the core state parameter of the battery model, which directly determines the precise

judgment of the battery management system on the remaining battery energy. Accurate SOC estimation can guide the optimization of charging and discharging strategies, avoid overcharging and overdischarging, ensure battery safety in use, and prolong the cycle life.

$$SOC(t) = SOC_0 - \frac{1}{C_n}\int_0^t \eta * I(\tau)d\tau$$

In the implementation of SOC estimation, we compared many algorithms and ultimately decided to adopt the **LSTM-Attention-AKF.**

The fitting effect is shown in the figure, where it can be seen that the estimation of SOC is extremely accurate at different time steps.

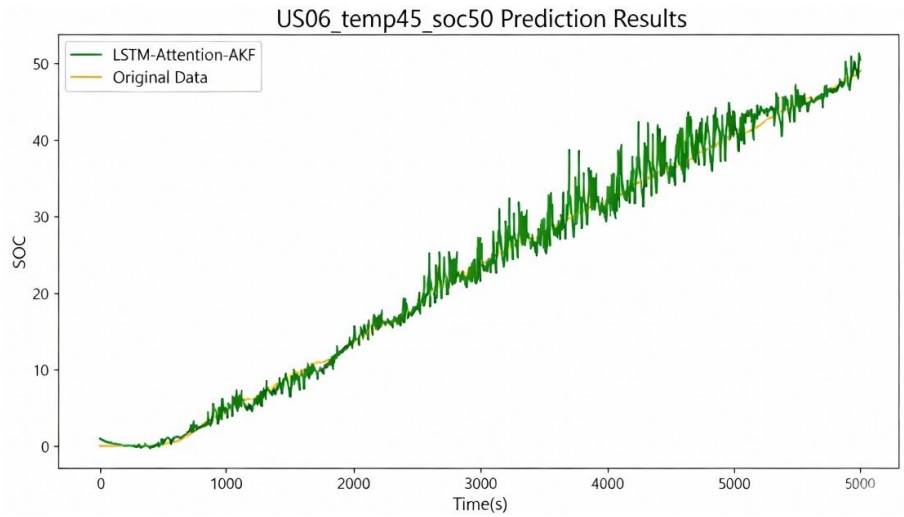

Figure 5: Analysis of SOC fitting effect

Verification 4: SOH

SOH is a core parameter for quantifying the degree of battery aging in battery models, directly reflecting the level of performance degradation after long-term use[2]. Reliable SOH data is a key basis for battery models to guide full lifecycle management and formulate maintenance or decommissioning plans.

$$SOH = \frac{C_{\max}}{C_n} * 100\%$$

Among them , $C_{\max}$ represents current actual maximum available capacity, C_n represents rated capacity of the battery at the time of manufacture.

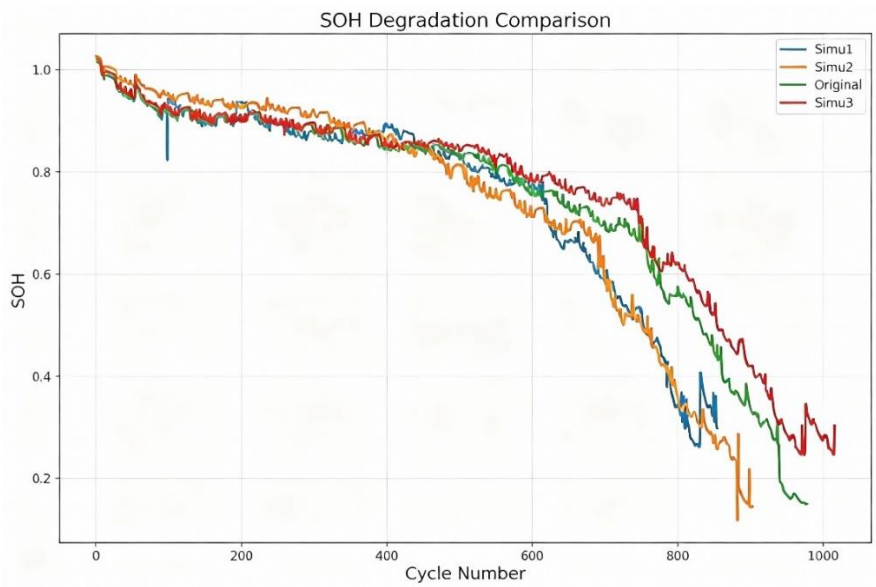

Figure 6: Analysis of SOH fitting effect

The fitting effect between our simulation results and actual data is shown in the figure. There are only parameter differences between different simulation groups, while the calculation logic remains the same. It is easy to see that the fitting effect of SOH is excellent.

4.5 Results

In task I, we successfully developed **a continuous time model THMSVM** to accurately predict the battery charge and discharge, and coupled external variables such as power consumption and ambient temperature of each module of the mobile phone into model THMSVM[3].

Based on all the above words, pictures and tables, we can draw the following conclusions:

A. SOH cannot be ignored, because the fitting effect of SOH varies greatly with the parameter settings, which reflects the complex electrochemical characteristics of the actual battery.

B. SOC is more sensitive to temperature than the initial estimate, which emphasizes the importance of temperature management in battery system design.

C. The polarization voltage and thermal voltage have a slightly larger error at the minimum and maximum of the time step, which is most likely due to the influence of temperature. Both the internal resistance and polarization resistance of the battery show a significant dependence on temperature, and the polarization capacitance increases with the increase of temperature. Therefore, how to cover the model fitting under the extreme temperatures on both sides is a challenge.

Therefore, the battery model we developed can be used for more accurate SOC estimation and battery management system design, and can also guide the battery use strategy.

5 Task II：Predicting Time-To-Empty

5.1 Problem Analysis

This task requires us to establish a model to predict the battery emptying time based on the THMSVM model and analyze the model performance, as shown in the following figure:

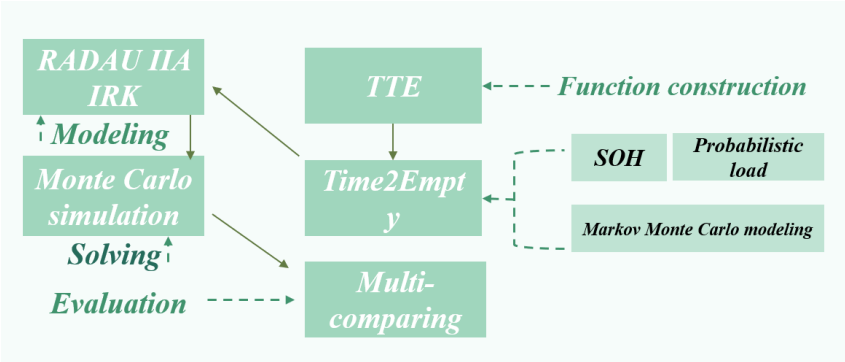

Figure 7: Thinking flow chart of task II

5.2 Establishment of the model

Part 1: Target function construction

Because the task requires the establishment of a battery prediction model, the objective function is naturally the depletion time TTE, which is defined as:

$$TTE = \min\{t \mid SOC(t) \le 0 \vee V_{sd}(t) \le V_{cf}\}$$

Although we designed the THMSVM model in the previous question, TTE still suffers from the randomness of the load (due to the high uncertainty of user behavior) and the time-varying state of battery health (SOH). Therefore, we need to add more parameters to enhance the confidence of the model's predictions.

Part 2: Markov Monte Carlo modeling

This model is designed for the "usage scenarios" of the problem, with six representative usage scenarios: Light Usage, Moderate Usage, Heavy Usage, Video Playback, Standby Mode, and Mixed Usage.

In determining the discrete Markov state space of a system,

$$S = \{1, 2, ..., M\}$$

The formula for estimating the one-step transition probability matrix is:

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

And P_{ij} satisfies $\sum_{j=1}^{M} P_{ij} = 1$.

Then we build the system evolution model under each state:

$$Y_{t+1}^{(i)} = f(Y_t^{(i)}, \varepsilon_t)$$

Finally, we conduct Monte Carlo random sampling and trajectory simulation, calculate

statistics for N simulated trajectories, and obtain the final output of the model.

Part 3: SOH modeling

As for the problem of battery health, the first question has already been explained. Here we only talk about the key points needed to build the model.

$$Q_m(SOH) = Q_v * SOH$$

The above formula is the formula of charge effective capacity attenuation with time. Although the attenuation of SOH can be approximately linear, it is obvious that the increase of internal resistance is the result of weighting of different modules, so it is nonlinear[4]:

$$R_{sum}(SOH, SOC, T) = R^*(T, SOC) * [1 + \ell(\frac{1}{SOH} - 1)]$$

We set the aging sensitivity coefficient to ℓ. when the internal resistance increases, the change of pressure drop is accelerated and non-linear and uniform.

Part 4: Probabilistic load modeling

Probabilistic load is rooted in the uncertainty of user behavior (such as screen brightness adjustment, network switching, application startup frequency, etc.), so it is necessary to build a probabilistic load model to accurately describe this feature:

$$\mathrm{I(t)} = \mathrm{I_b} + \sum_{\mathrm{m \in M(t)}} \mathrm{I_m} + \epsilon(\mathrm{t})$$

We might as well assume that the current follows a lognormal distribution, because the peak value of users' power consumption cannot always appear at noon. Considering users' sleep in the morning, the peak value of the possible normal distribution may be slightly skewed to the right:

$$\mathrm{I_m} \sim \mathrm{N}(\mu_\mathrm{m}, (\mu_\mathrm{m} \cdot \sigma)^2), \mathrm{I_m} = max\ (0.001, \mathrm{I_m})$$

After finishing the above work, we focus on the user himself. The user behavior shows a certain rule in the time dimension, and these discrete data are collected and aggregated into set A:

$$A(t) = \{A_h \,|\, t \in [t_0, t_1]\}$$

5.3 Solving model

Part 1: Radau IIA implicit Runge Kutta method

For the stiff ordinary differential equation of SOC evolution, the **Radau IIA method** has the characteristics of a - stability and even L - stability, and its stability is far better than the traditional explicit Runge Kutta method, especially suitable for large-scale simulation scenarios such as SOC cycle life prediction, so we choose this method rather than the traditional Runge Kutta method.

We need to substitute Radau IIA iterative formula to build implicit equations:

$$k_i = f(t_n + c_i h, y_n + h\sum_{j=1}^{s} a_{ij} k_j)$$

Then we use Newton Raphson method to solve the above implicit equations, give the initial guess value of k_i, and iterate until the residual meets the accuracy threshold to obtain

all k_i.

Finally, we update the SOC numerical solution to get the SOC value at the next time:

Sensitivity Analysis:

$$y_{n+1} = y_n + h\sum_{i=1}^{s} b_i k_i$$

Part 2: Uncertainty quantification

The uncertainty of battery SOC prediction comes from a wide range of sources. The user differences in battery aging degree, the unrepeatability and unpredictability of user behavior may lead to SOC fluctuations.

In order to quantify these uncertainties systematically, the Monte Carlo simulation (MCS) method is adopted, the simulation times are N=200 (in pursuit of calculation accuracy and quantization degree), and the following parameters are randomly sampled for each simulation: $T, SOH, I(R)$.

$$tr_p(t) = \sum_{i=1}^{N} (\{tr_i(t), p\})$$

And $p \in \{5, 50, 95\}$, the 90% confidence interval is constructed through the three values of P, which can effectively reflect the SOC change under different intervals, and quantify the uncertainty index, which improves the accuracy of the model.

5.4 Model index validation

In addition to the different SOC and usage scenarios required by the topic, we have added many additional items to verify the performance of the model.

We divide them into two categories, which are shown in the following figure:

Verification 1: Temperature influence

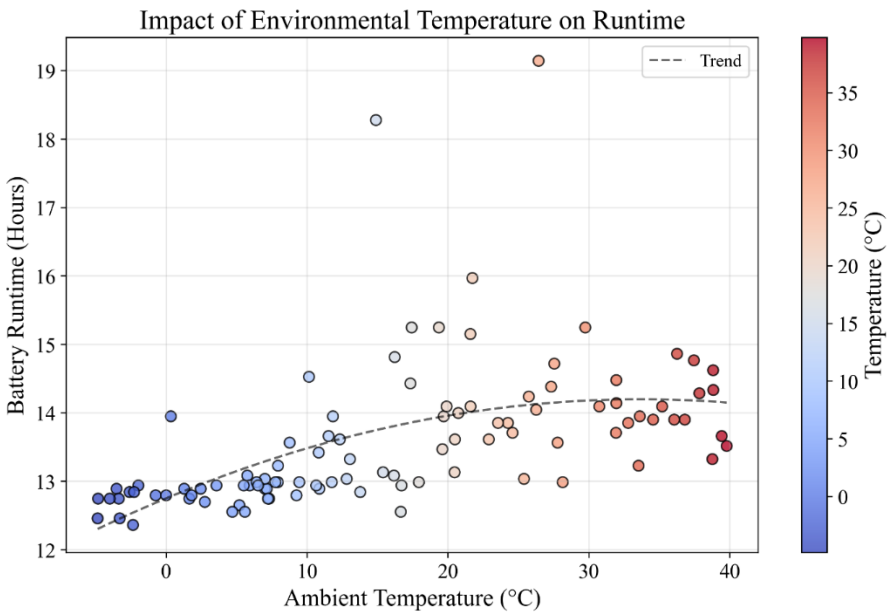

Figure 8: Influence of ambient temperature on evacuation time

It can be seen from the figure that 20-35 ℃ is the optimal working range of the battery,

and the capacity and efficiency are at the peak; SOC and internal resistance changes at other temperatures lead to endurance attenuation.

Verification 2: Usage scenarios influence

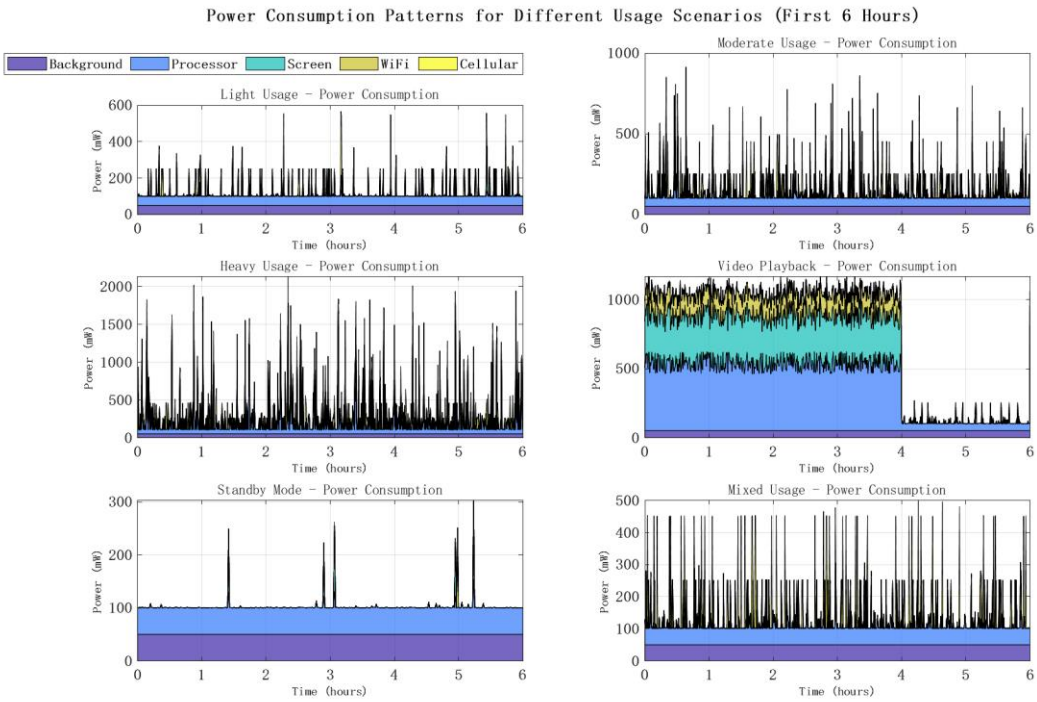

Figure 9: Influence of different Usage scenarios

The figure shows the power consumption of different modules under 6-hour use in different scenes[5]. It is easy to conclude that the power and power fluctuation of heavy use are the largest, the standby mode is the smallest, and the power consumption of video playback scenes is relatively stable.

Verification 3: SOC

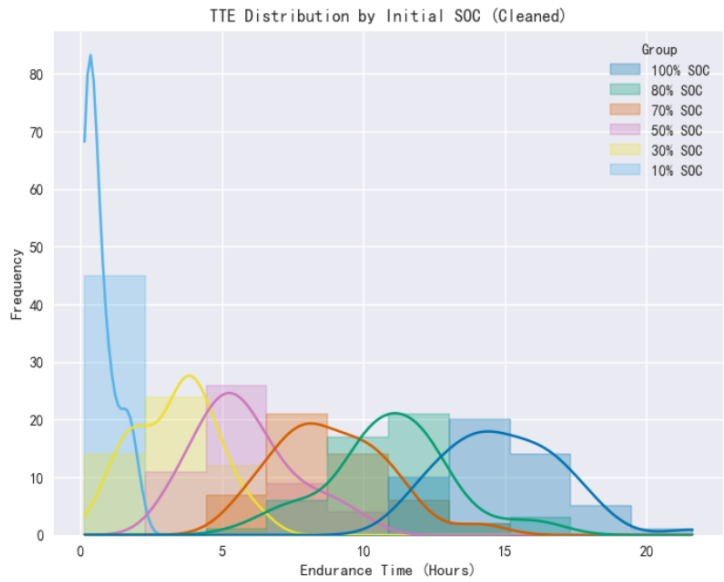

Figure 10: Influence of different SOC

The difference in initial SOC directly determines the length of TTE, and the smaller the SOC, the earlier the peak appears, the shorter the maintenance time, and the shorter the running time, the more variance the Markov random process accumulates, thereby reducing the standard deviation.

Verification 4: SOC & Usage scenarios

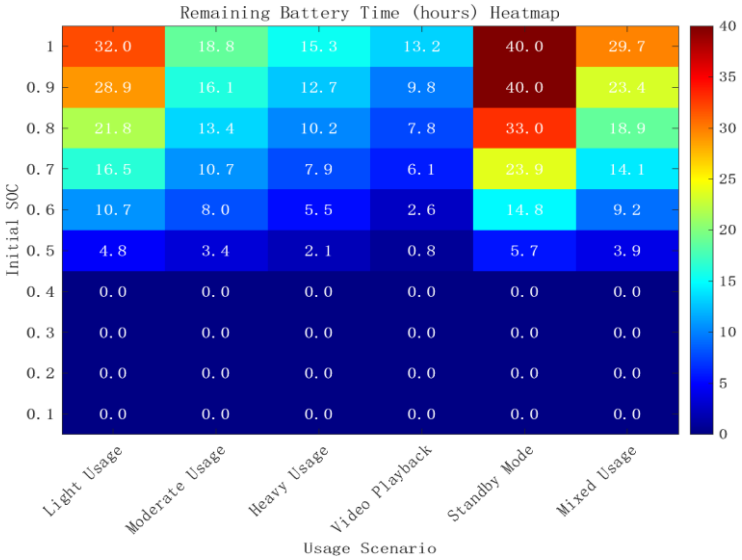

Figure 11: Influence of different SOC & Usage scenarios

From the heatmap, it can be observed that even in heavy usage scenarios, the SOC of the battery is high and the remaining time is still very long.

Verification 5: SOH

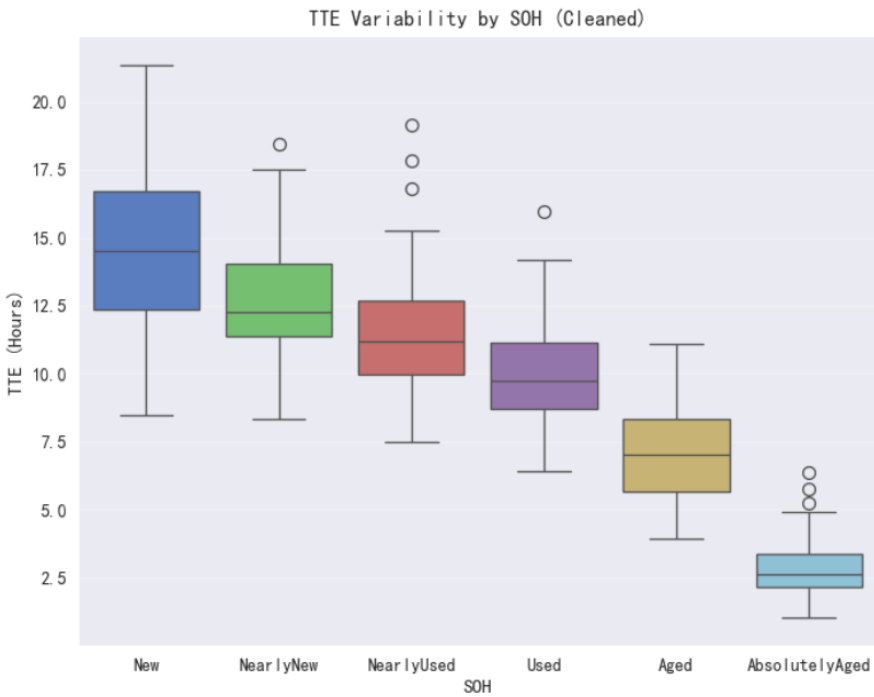

Figure 12: Influence of different SOH

As shown in the figure, the impact of SOH is not just a simple attenuation of TTE. From

full charge to 70% charge, the decrease in TTE is greater than 30%, exceeding the attenuation of SOC itself. This is due to the parasitic effect of increased internal resistance of SOH attenuation.

5.5 Results

A: We compare the influencing factors horizontally:

Table 2: Comparison of different factors

Factors	$\overline{TTE}$	$\triangle TTE$ (%)	Influence
Usage scenarios	$17.83 \rightarrow 9.24$	-48.18%	High
SOH	$13.83 \rightarrow 5.94$	-57.05%	High
SOC	$15.27 \rightarrow 11.76$	-22.99%	Middle
High T	$15.98 \rightarrow 13.89$	-13.08%	Middle
Low T	$15.98 \rightarrow 14.67$	-8.20%	Low

B: We also analyzed the combination of these factors, and the activity that led to the **most reduction** of battery life was the **continuous use of high brightness screen in low temperature** environment; **GPS module, standby use scenarios** and other factors have surprisingly **little impact** on the model, which is consistent with the common sense in life that "the screen consumes the fastest power, and the low-temperature endurance shrinks".

6 Sensitivity test and analysis

6.1 Sensitivity test modeling

We use the **single factor perturbation method (OAT)** for local sensitivity analysis, while the global sensitivity is realized by the **Latin hypercube sampling (LHS) hybrid random forest model**, which represents the contribution ratio of all parameters in the whole space.

Stage 1: Local sensitivity analysis

We calculate the reference output $y^* = f(x^*)$ by setting the reference value $x^* = (x_1^*, x_2^*, ..., x_n^*)$ of each parameter.

Then apply x_i small disturbance Δx_i to a and calculate the output after disturbance:

$$y_i^+ = f(x_1^*, ..., x_i^* + \triangle x_i, ..., x_n^*)$$

After that, continue to calculate the sensitivity coefficient Si:

$$Si = \frac{\partial y}{\partial x_i} * \frac{x_i^*}{y^*}$$

Lastly, it is determined that if $|Si|$ is larger, the local influence of parameter x_i on output y is more significant.

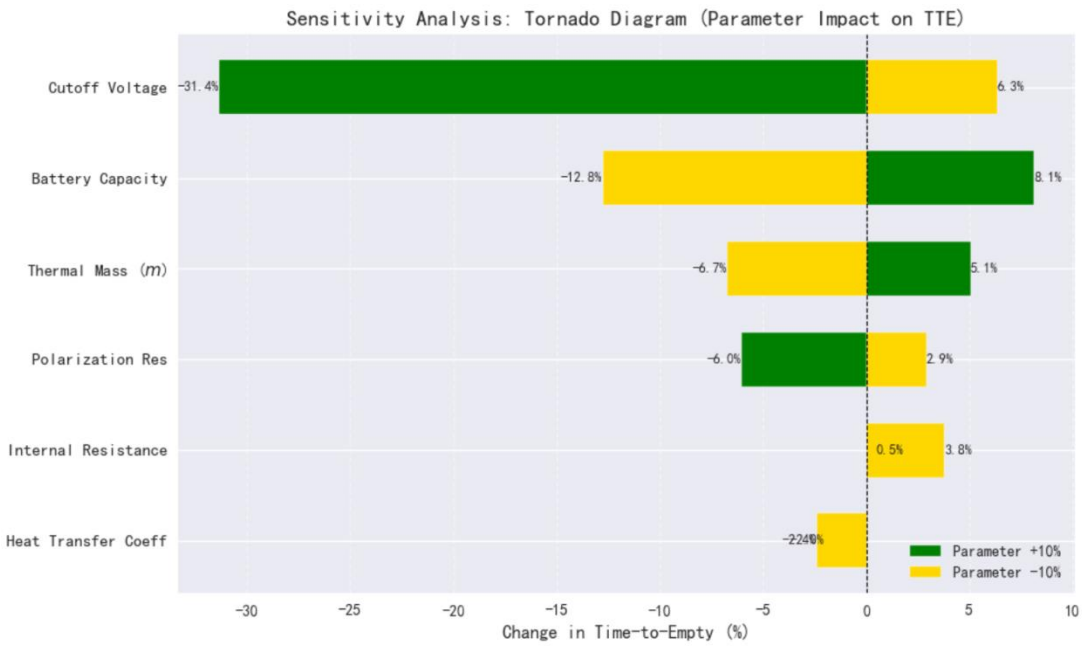

Figure 13: Local sensitivity analysis

Stage 2: Global sensitivity analysis

We use Latin hypercube sampling (LHS) to efficiently cover the whole space of parameters to generate samples, use random forest (RF) to fit the input-output mapping relationship, and quantify the global sensitivity of parameters based on the importance of RF characteristics.

The nonlinear mapping relationship between fitting input and output should be:

$$\hat{y} = f(x_1, x_2, ..., x_n)$$

The calculation of mean square error reduction (MDA) meets the following requirements:

$$I_i = \frac{1}{M} \sum_{m=1}^{M} (MSE_m^{perm(i)} - MSE_m)$$

Where $MSE_m^{perm(i)}$ is the prediction error after replacing X_i, and the global sensitivity coefficient is normalized to:

$$S_i = \frac{I_i}{\sum_{j=1}^{n} I_j} X_i$$

The analysis results are shown in the table below:

Table 3: Global sensitivity analysis

Rank	Parameter	Importance_mean	Importance_std
6	C	0.773466	0.070736
5	V_{cutoff}	0.607517	0.052738
4	R_{pol}	0.117675	0.008349
3	K_{heat}	0.116933	0.008647

2	M_{heat}	0.111415	0.005885
1	R	0.107392	0.005072

In the global sensitivity test, there is an abnormal phenomenon compared with the local sensitivity test: the importance of capacitance exceeds the cut-off voltage.

This is because in the local oat analysis, if the local reference range is at the beginning and end of the voltage change curve, it only needs a small disturbance to make the TTE change violently. Therefore, the global sensitivity test is more universal and universal for global search.

6.2 Sensitivity test analysis

Part 1: Modeling assumptions fluctuation

The variation factors of modeling assumptions are extremely wide, and the small disturbance of all parameters in the previous discussion may cause great changes in sensitivity and prediction results.

In order to better study the impact of modeling assumptions, we discussed three different situations that are frequently used in common situations. They are respectively marked as **Light User**, **Heavy User** and **Random User (Normal)**.

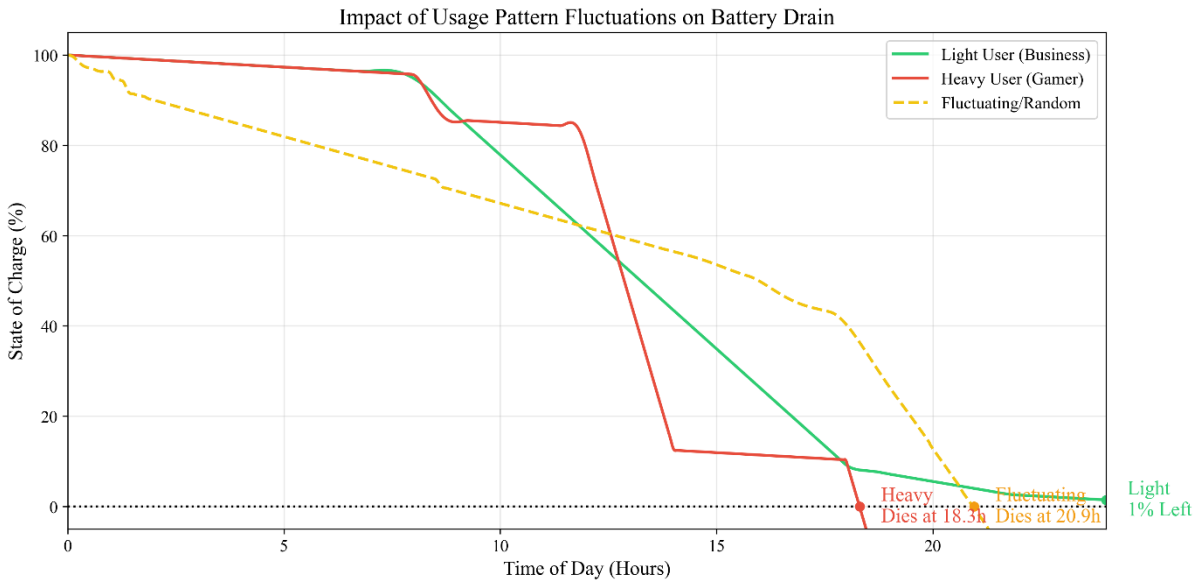

Figure 14: TTE changes caused by modeling assumptions

It can be seen from the figure that the curve alignment of the three modeling assumptions conforms to the common sense of the real world, which proves the accuracy of our modeling parameter settings in extreme cases.

For the "heavy use" modeling assumption, we introduce the component interaction loss and Arrhenius temperature assumption. The power consumption of different modules is not independent, which better simulates the phenomenon of the physical world.

Part 2: Parameter value fluctuation

Because the sensitivity and prediction changes caused by parameter value fluctuations are divided into single parameter and multi parameter mixed effects, we consider the mixed pa-

rameter interweaving effects closer to real life, and the coupling parameters are screen brightness and network.

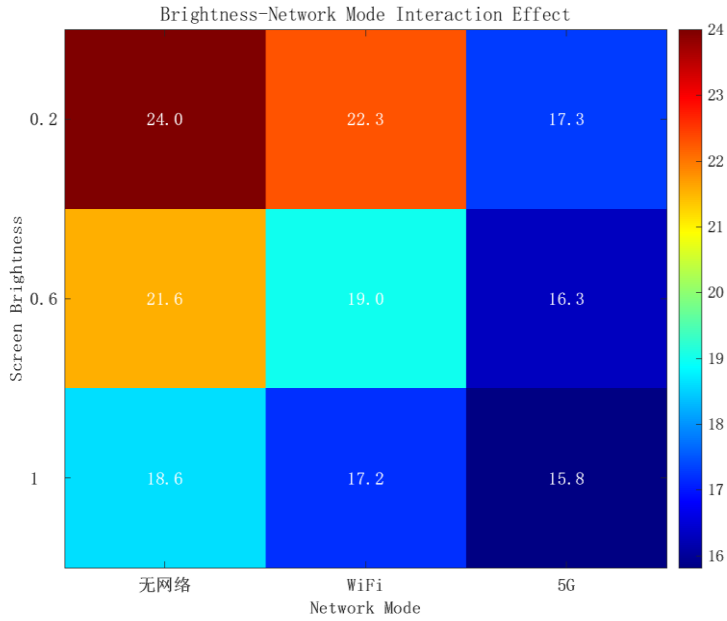

Figure 15: Influence of parameter interleaving and mixing on TTE heat map

This network brightness coupled parameter heat map can more universally reflect the combined effect of the two parameters. Obviously, it is not a simple linear superposition, but there is a corresponding interaction.

When the power consumption of the device network is large, the marginal effect of the power consumption components superimposed due to brightness on TTE is actually reduced, so we can appropriately reduce the brightness of the screen when using the high energy consumption network to improve the SOH and instant performance of the battery.

Part 3: Usage mode fluctuation

The usage mode has been described in detail above, and the definition of classes will not be explained here.

Here, we will only discuss the impact of battery TTE caused by the use mode switching due to the randomness and unpredictability of user behavior.

Table 4: Comparison of different usage scenarios

Usage scenarios	Depletion time(h)	Usage scenarios	Depletion time(h)
Light Usage	25.1	Video Usage	18.5
Moderate Usage	20.8	Standby Mode	30.9
Heavy Usage	18.3	Others	18.4

It is not difficult to see from the figure that the "heavily used" battery has the shortest depletion time because it continuously activates high-power components; The battery depletion rate of "Others" users is also very fast, mainly because of the additional power consumption caused by frequent switching of different use modes.

6.3 Results

After the global and local sensitivity tests and the predicted changes under three conditions, we can get:

A: Influence ranking and reasons

(1)The most influential C & V_{cutoff} The former determines the maximum depletion time in the physical model, and the cut-off voltage determines the depletion time, thus controlling the actual utilization efficiency of the battery.

(2)Influential R & R_{pol} Core components of ohmic heating

(3)Less influential M_{heat} & K_{heat} In non extreme cases, thermal parameters can be regarded as constants with little influence.

B: Impact of different fluctuations

Table 5: Impact under different fluctuations

Variable	Impact
Modeling assumptions	Accelerating the arrival of depletion time if extreme use
Parameter value	Lowing the marginal benefit of battery TTE while interacting
Usage mode	Frequent switching mode damages SOH

7 Suggestions and promotion

7.1 Client

Obviously, it is not specific enough to make some well-known suggestions in the macro view, so we quantify the user behavior by introducing several models.

User behavior is determined by component power consumption, but different from the above, we have added mode operator and temperature operator here.

The mode operator is used to better characterize the power (lower energy consumption) in standby mode. The temperature operator is used to characterize the change of the additional term ΔT caused by R with the increase of temperature.

$$\begin{cases} P_p(t) = P_{sc}(t) + \mathrm{P}_{rf}(t) + P_{cpu}(t) + P_o \\ P_{bv}^* = \alpha_{bv} * P_{bv} \rightarrow O\text{perator}M \\ R^*(t) = R(T_0, t, \Delta T, SOC) \rightarrow O\text{perator}T \end{cases}$$

Through the above model, we quantify the different behaviors of users as follows:

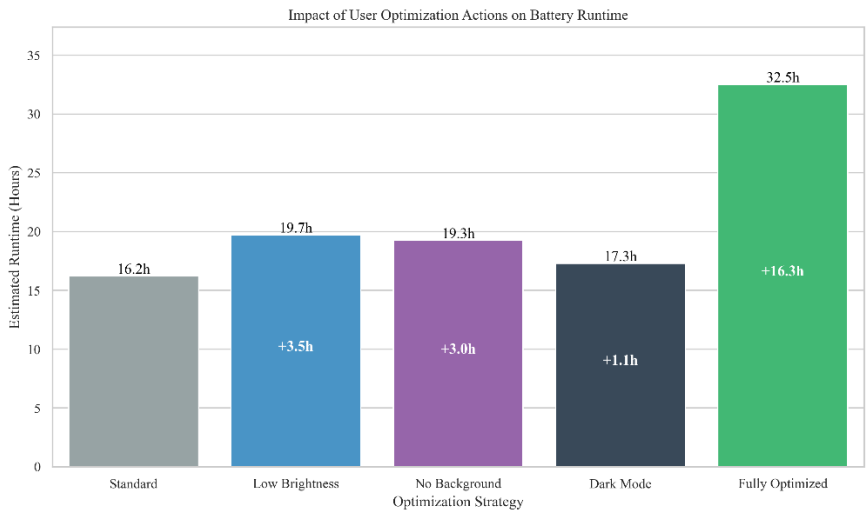

Figure 16: Quantitative optimization of user behavior

t can be seen from the pictures that users can optimize the battery life by **reducing screen brightness (the most effective)**, **disabling background tasks (the second effective)**, **switching network mode (WIFI24G)**, **avoiding low temperature environment** and other operations. If conditions permit, it is better to combine the above operations.

7.2 Operating system side

What developers need to consider can be modeled as a defensive security model. The ultimate goal is to ensure that the system is normal and that the power margin is as large as possible to meet the corresponding needs of users.

$$\begin{cases} P_m(t) = I_{\mathrm{m}}(t) * V_{cf} \\ P_{margin}(t) = \min(\mathrm{P}_{ref}, P_m(t)) \end{cases}$$

The optimization of operating system is the most direct and effective for developers from the perspective of module design of battery and battery management system. Therefore, we study the power consumption optimization of different modules.

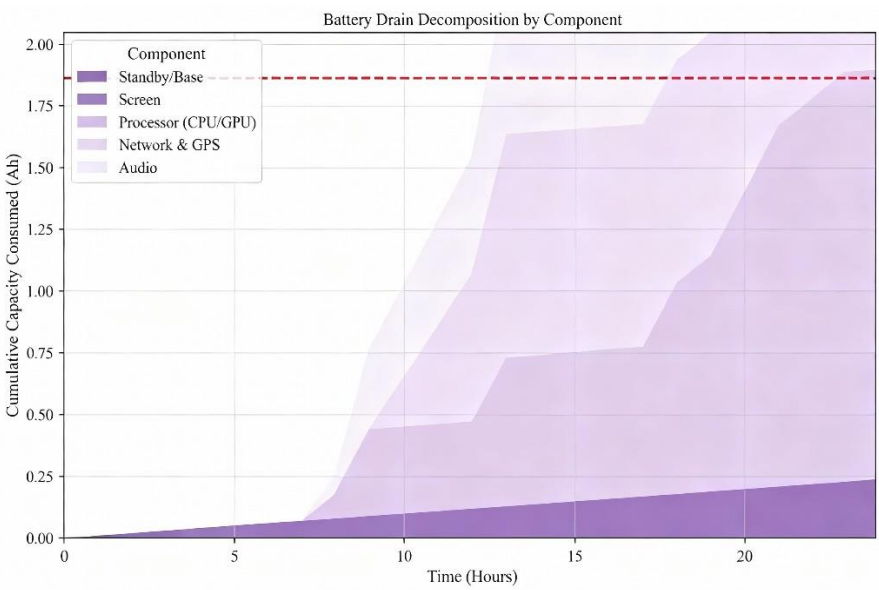

Figure 17: Power consumption contribution of different modules

It is not difficult to see from the figure that the power consumption of the processor accounts for a large part, so developers can try to **dynamically adjust the internal resistance technology**, using MOSFET instead of pure resistance.

In addition, different modules do not always need to work in parallel. It is also important to intelligently monitor the user's behavior of using mobile phones and **automatically close the background applications.**

7.3 Popularizing

Part 1: Effective capacity reduction induced by battery aging

The concept of SOH has been explained in detail before and will not be repeated here. Here we analyze the influence of SOH on the effective capacity of battery from the perspective of thermoelectric coupling.

Firstly, battery aging (such as cyclic charge and discharge, storage attenuation and other behaviors) will lead to **internal active material loss**, **electrolyte decomposition**, and **electrode structure damage**, which will directly reduce the total amount of charges that can be stored and released, that is, the effective capacity.

Then, SOH is defined by the ratio definition formula: $SOH = \frac{C_{actual}}{C_{standard}}$.

When SOH decays and $C_{standard}$ remains unchanged, C_{actual} naturally decreases to ensure the change of SOH.

Part 2: Deeper applications

There is an ancient Chinese saying that goes, **'The great road leads to simplicity.'** It teaches us to analyze problems by examining their internal logic and essence.

Essentially, our model can be described using vector combinations:

$$V = [Equations, Constraints, InterdisciplinaryPoints]$$

In the construction of the battery model, $InterdisciplinaryPoints$ is divided into $Thermal$ and $Electronic$. Therefore, any field that involves the intersection of thermodynamics and electrical coupling can achieve model transfer.

This has broad prospects in the portable device market, such as **flexible smart tight fitting clothes,** which monitor the thermodynamic parameter of human body temperature and convert it into readable electrical signals, which are sent to terminals to achieve the loosening or tightening of clothes.

Alternatively, it could be a **non-invasive continuous blood pressure monitor** that migrates the equivalent circuit model (ECM) to the electrode skin interaction system to calibrate signals such as pulse waves to reduce distortion; Establish a sensor power consumption heating blood pressure error feedback chain using a thermoelectric coupling model to compensate for temperature interference in measurement.

Or for **smart rings**, the electrode skin contact impedance can be calibrated using an electrochemical impedance spectroscopy (EIS) model to improve the accuracy of physiological

signals; The micro thermoelectric coupling model reduces the interference of sensor heating on body temperature measurement, ultimately achieving the unity of seamless wearing and precise monitoring.

8 Strengths and Weaknesses

8.1 Strengths

● Balance: Our model achieves a perfect balance between resources and performance. THMVM model couples multidimensional vectors on the basis of the first-order Thevenin model, making it easy to reproduce and commercially use in engineering. Compared to the more novel ESPM extended model and Improved solid state P2D model models in academia, our model consumes much less computational resources.

● Flexibility: Because the model is relatively simple and does not involve special reactions in the field of electrochemistry, it has strong portability and is easy to be transplanted to other non-battery applications.

● Stability: After global sensitivity analysis and local sensitivity test, our model is very stable in a large voltage and temperature range.

● Scalability: Our model has been verified by NASA data set, battery data from University of Maryland, and data set on Kaggle website, and the fitting effect is excellent.

8.2 Weaknesses

When extreme conditions (temperature, voltage) are involved, the fitting effect of our model is still different, but this problem can be solved by circuit structure design, dynamic parameter adjustment, etc.

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Report on Use of AI Tools

1. Doubao(Dec 18, 2025 version)
   Query: Understand the types of battery models in industry and academia
   Output: Second order RC, Thevenin, Electrochemical model, Machine learning aided model
2. Google Translate (Jan 23, 2026 version)
   Translate some academic Chinese into English
3. Gemini3 (2025 version)
   Query: How to conduct battery sensitivity analysis
   Output: Global and local analysis, Monte Carlo analysis. Clarify the analysis objectives and core parameters, and determine which parameters may affect the results. Set the change range of parameters and keep other parameters unchanged. Adjust the parameters one by one and calculate the changes of the corresponding target results. Comparing the fluctuation range of the results under the change of each parameter, the parameter with large fluctuation is the sensitive factor. Summarize sensitive factors and give suggestions.
4. Github Copilot (Jan 16, 2024 Version)
   Complete the code of PSO algorithm
5. KIMI (Jan 27, 2026 Version)
   Query: How to extend the battery model to other portable devices?
   Output: Maintain the core framework of continuous time dynamics, find equipment with thermoelectric coupling effect at the same time, and re calibrate the component power consumption coefficient, temperature correction operator and battery attenuation factor for notebook computers, smart watches and other equipment;